

IIoT (Industrial Internet of Things) Communication Protocols

A Comparison of MQTT, OPC UA, and CoAP

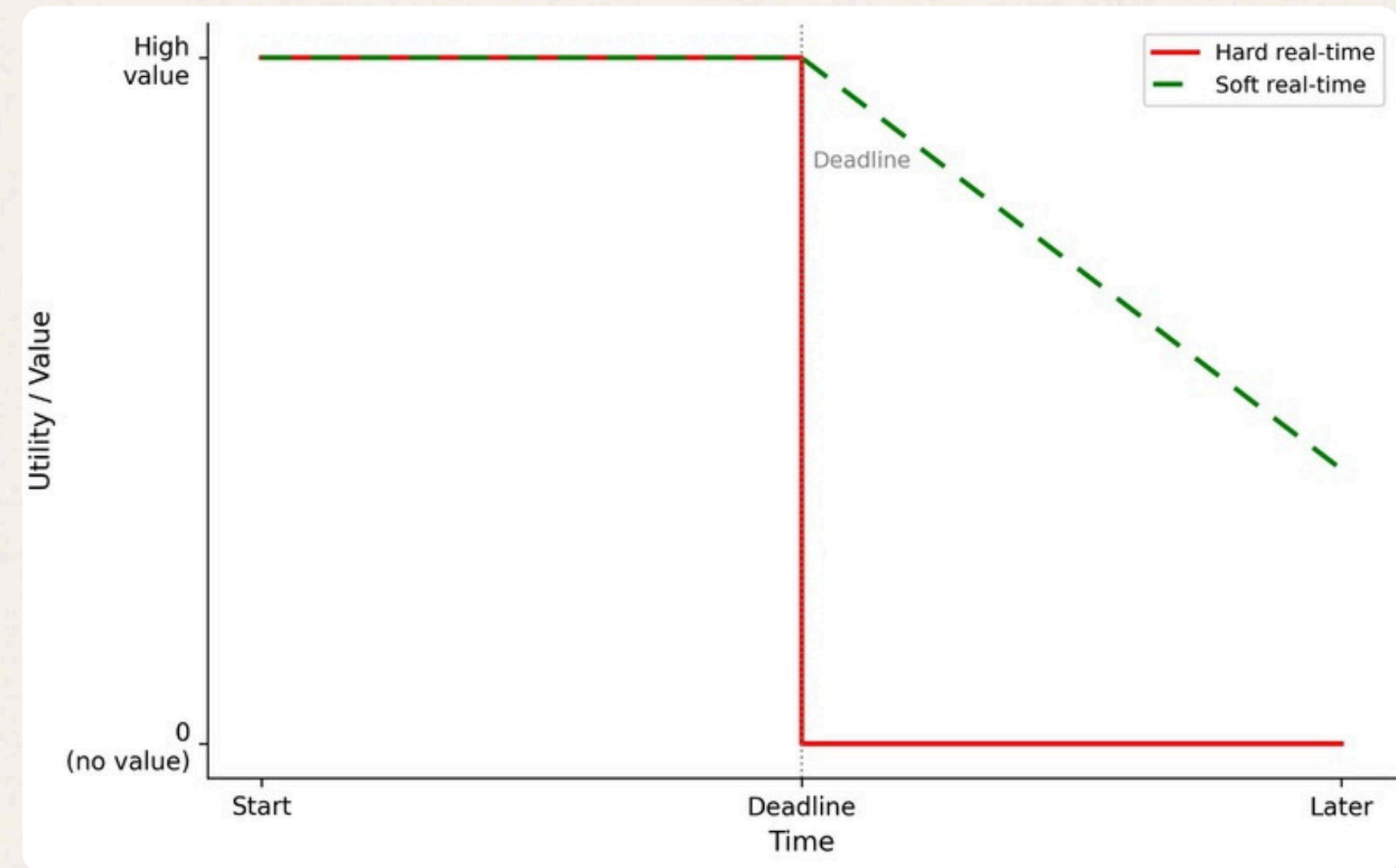
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Industry Needs Specialized Communication

- Demand strict real-time performance and high reliability
- Must handle mixed traffic by importance: control loops, sensor data, configuration info
- Unlike "best-effort" Internet protocols, industrial standards guarantee reliable, consistent data sharing
- Works across different devices and manufacturers even when things go wrong

Technical Requirements: Real-Time & Low Latency⁰³

- Processing within a precise, predictable time frame
- Key for robotics & automation
- Hard: missed deadline = failure (medical, nuclear)
- Soft: missed deadline = lower quality (video)
- Latency: ~1–10 ms
- Goal: deterministic, no data loss



Technical Requirements

- **Security:** more connected devices = larger attack surface
- **Risks:** process disruption, IP theft, loss of company data, business espionage
- **Cybersecurity:** remains a major obstacle to full IIoT adoption
- **Scalability:** rapid device growth must scale horizontally without performance loss
- **Interoperability:** RAMI 4.0 requires vendor-independent communication at every level

Operational Requirements:

- Technical specs alone aren't enough
systems must be dependable in operation
- RAMS(S) defines five pillars of industrial dependability →
- Originally RAMS; Security (S) added for the connected era
- Each pillar maps to a real-world failure risk

Letter	Property	Description
R	Reliability	System performs required function without failure over specified time
A	Availability	System is accessible and operational when required
M	Maintainability	System can be restored to operational state quickly after failure
S	Safety	No harm to people, equipment, or environment on failure
(S)	Security	Protection against unauthorized access and cyber threats

Communication Protocols

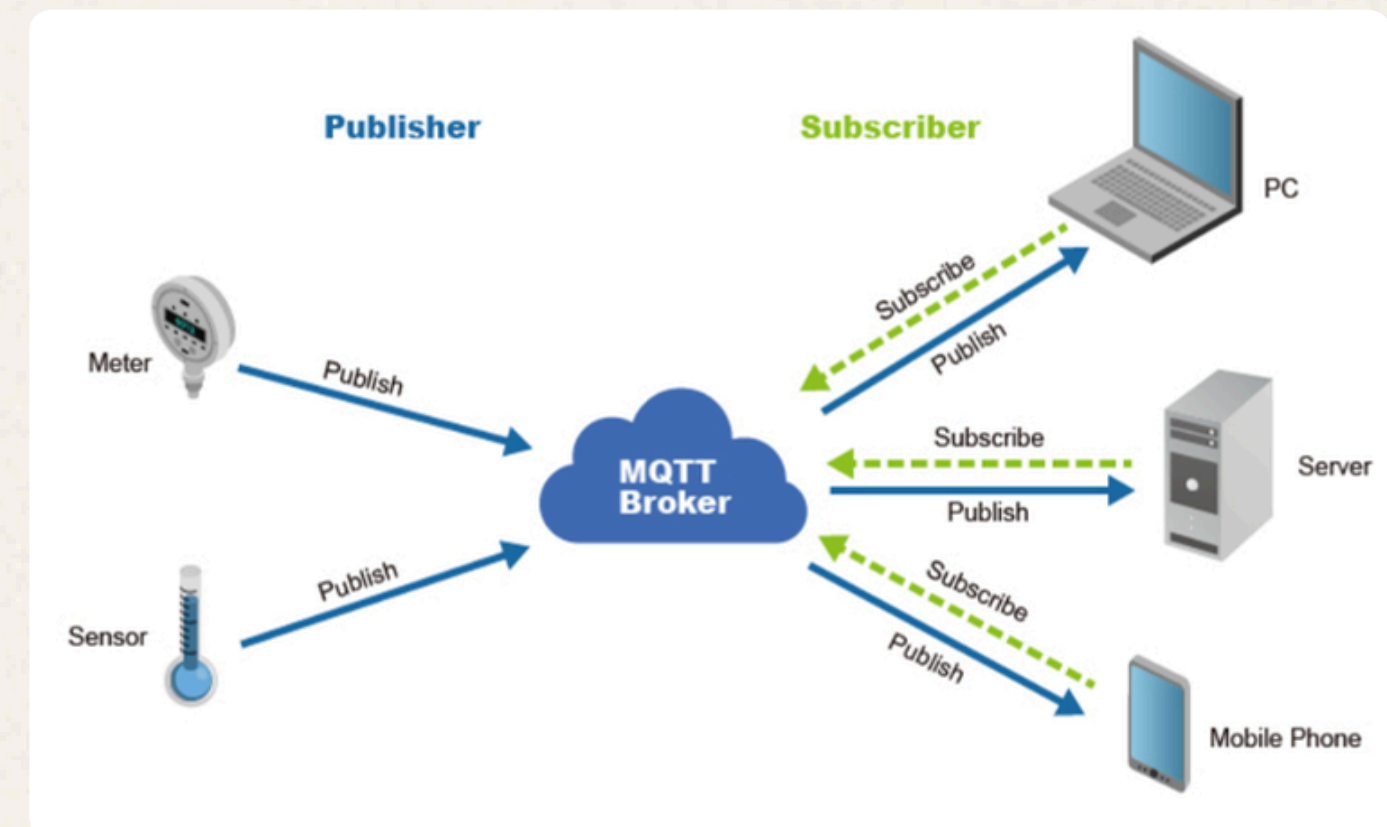
- **MQTT** : Message Queuing Telemetry Transport
- **OPC UA** : Open Platform Communications Unified Architecture
- **CoAP** : Constrained Application Protocol
- **Modbus** : Modified Bus
- **HTTP** : Hypertext Transfer Protocol
- **AMQP** : Advanced Message Queuing Protocol

MQTT (Message Queuing Telemetry Transport)

07

Lightweight publish-subscribe protocol for reliable IIoT messaging

- Origin: IBM, 1999 (pipeline monitoring) → OASIS standard 2014 → MQTT 5.0 (2019)
- Publish-subscribe model: Publisher → Broker → Subscriber
- Decoupled design: peers need not know each other or be online together ideal for unreliable networks
- Lightweight: runs over TCP/IP, just a 2-byte header (ports 1883 / 8883 for TLS)
- 3 Quality-of-Service levels: QoS 0 (at most once), QoS 1 (at least once), QoS 2 (exactly once)
- Key features: Last Will & Testament, Retained Messages, Keep-Alive

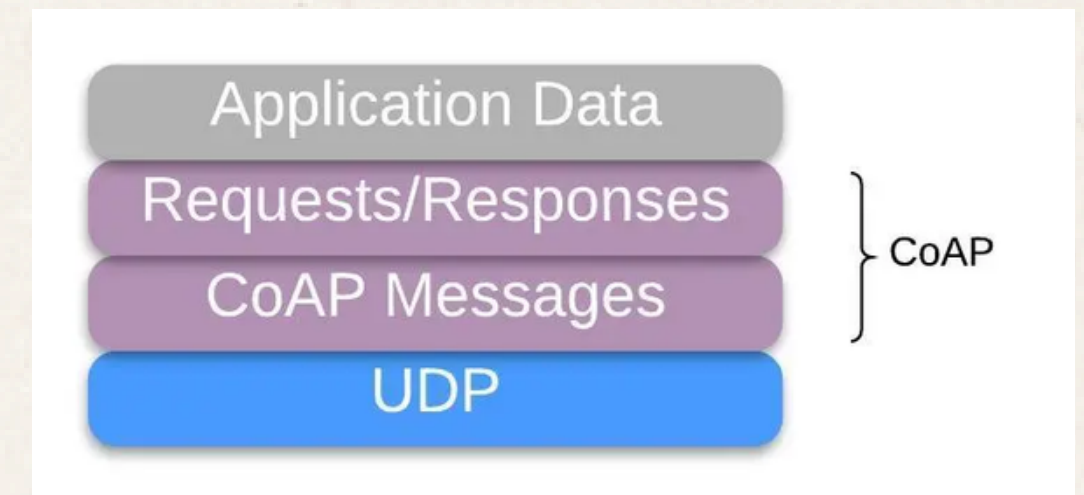


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Constrained Application Protocol - CoAP

08

- Target Devices: resource-constrained Class-1 on low bandwidth networks
- Architecture: REST architecture with GET, PUT, POST, and DELETE methods built on UDP. extremely small size
- Push Model: Asynchronous observer pattern, resources push notifications when data changes
- Web Interoperability: Easily translationable between CoAP and HTTP without application knowledge
- known as “Poor mans HTTP”

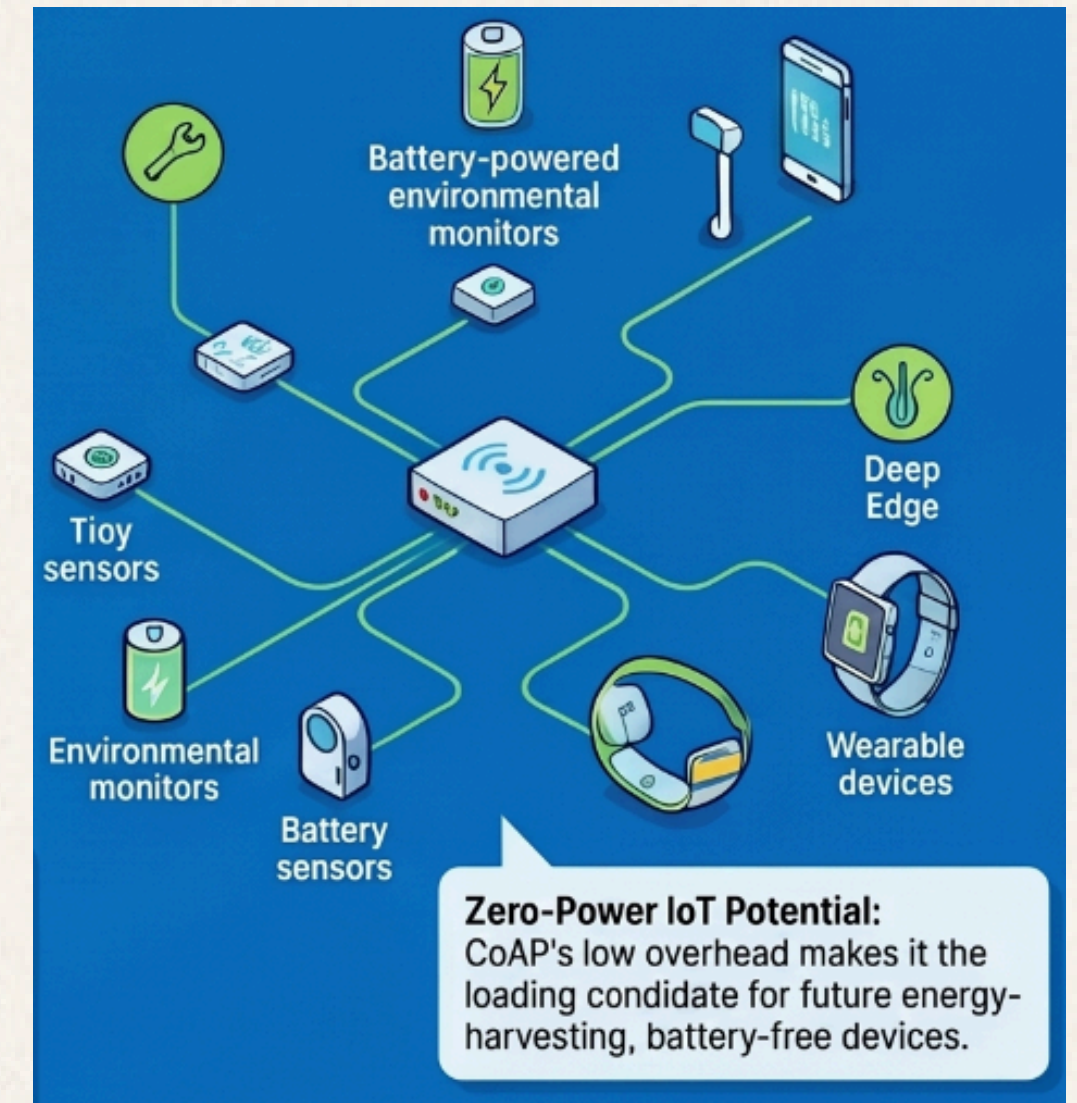


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CoAP in IIOT context

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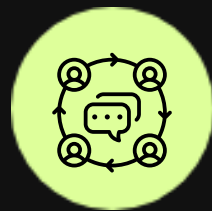
- Low Overhead & Bandwidth: With a compact *4-byte* header
- Built-in Reliability: operates over UDP, low overhead compared to TCP.
- Resource Observation: "Observe" mechanism, client to subscribe to a sensor(event-driven).
- Multicast Support: Can send a single message to multiple devices simultaneously.
- Web Integration: can be easily linked to enterprise cloud dashboards or legacy SCADA systems.



[3]

Open Platform Communications Unified Architecture (OPC UA)

OPC UA is a secure and interoperable communication standard that enables seamless data exchange between devices, systems, and diverse industrial automation networks, serving as a key enabler of IIoT and cross-network integration.



Flexible Communication

Client-server & publish/subscribe communication using TCP-based and HTTP/XML protocols.



Real-Time Sync

Advanced data modeling integrating with PROFINET & EtherCAT for time-critical networks

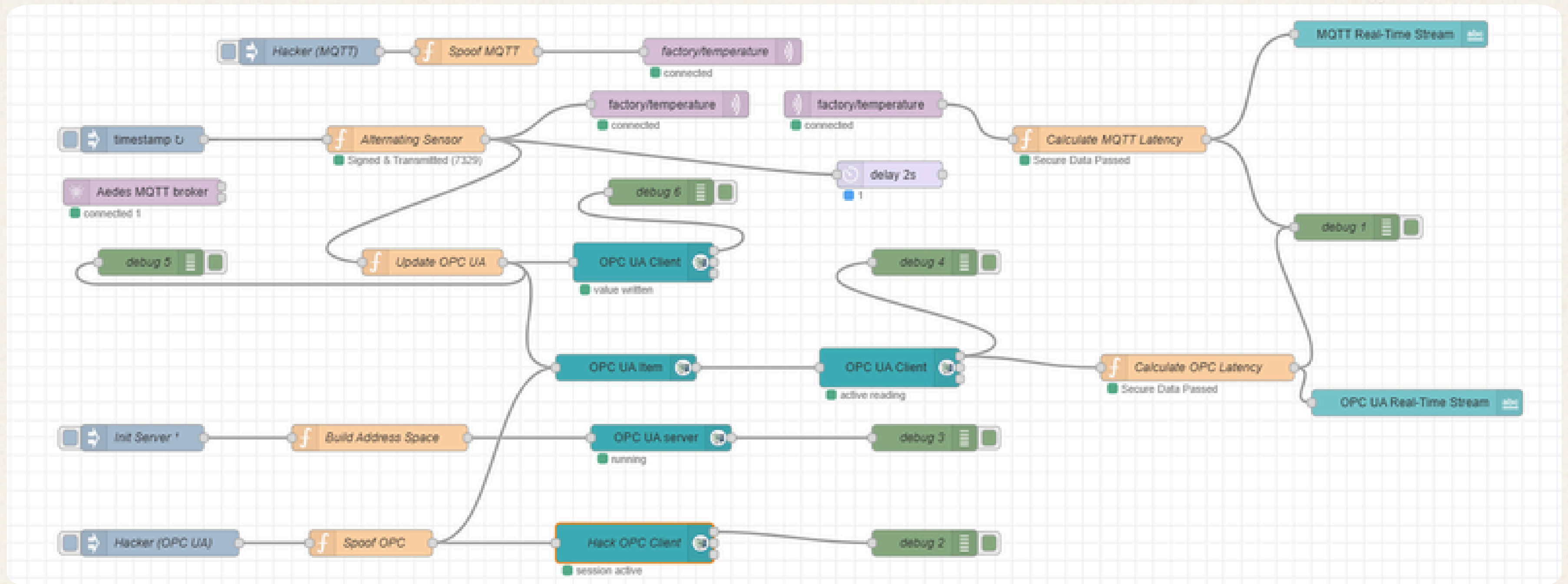


Secure Auto-Setup

Plug-and-Produce (PnP) discovery protected by X.509 certificates & robust encryption.

Case Study : MQTT vs OPC UA

11



Case Study : MQTT vs OPC UA

- Faster communication in MQTT than OPC UA
- MQTT is significantly more suitable for bandwidth-limited and high-frequency communication environments (**MQTT-132 Bytes, OPC UA - 44 to 236 Bytes**).
- Transport Layer Defense
- Application Layer Validation

Protocol Monitor

OPC UA Real-Time Stream

Latency: 8 ms | Data: 81.90°C | 
VERIFIED

MQTT Real-Time Stream

Latency: 5 ms | Data: 81.90°C | 
VERIFIED

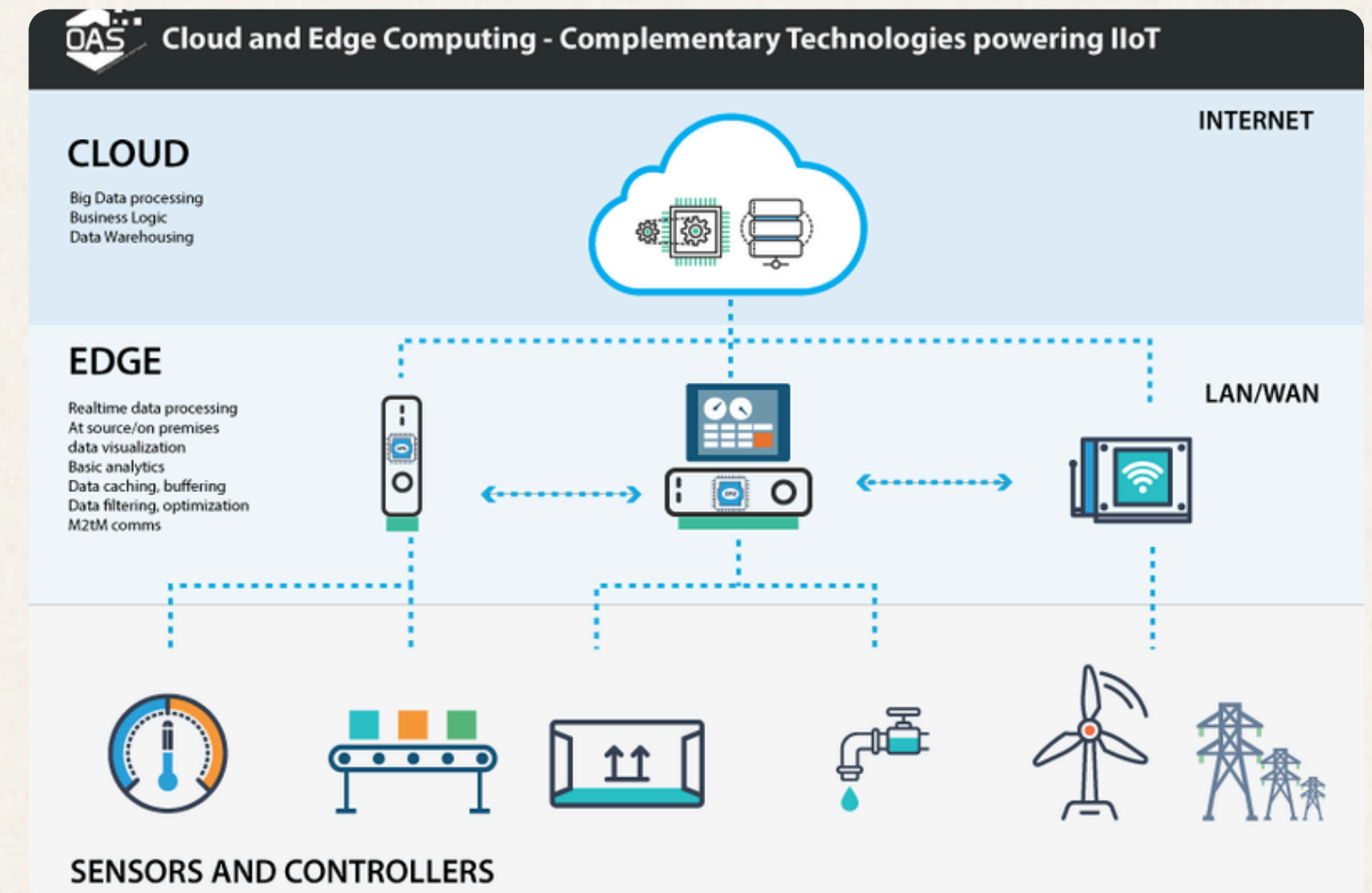
Feature	MQTT	OPC UA
Model	Publish-Subscribe	Client-Server
Components	Broker, Publisher, Subscriber	Server, Read/Write Clients, Address Space
Data Organization	Basic topic-based routing	Advanced semantic structured nodes
Deployment	Simple, lightweight, fast	Complex, structured, highly organized

DATA ACQUISITION, MONITORING, AND CLOUD INTEGRATION

In modern IIoT systems, communication protocols enable machine-to-machine communication within factories and support data transfer from devices to edge and cloud platforms for monitoring, analysis, and decision-making.

Device → Edge Gateway → Cloud Platform

- MQTT enables lightweight edge-to-cloud data transfer
- CoAP supports low-power sensor communication
- OPC UA connects devices with SCADA/MES systems
- OPC UA PubSub with MQTT enables secure cloud integration

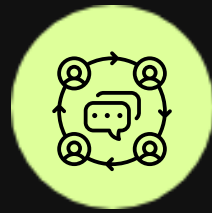


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Current Challenges: Protocol Analysis

14

Reliability Gap: Industrial automation demands 99.99% uptime and <100ms latency, currently hindered by co-existence issues and complex spectrum sensing.



MQTT

- Lacks in-built application-layer cryptography.
- Access control often relies on default logins.
- High packet loss under heavy parallel loads.



CoAP

- Traditional TLS is impractical for constrained devices.
- Setup mechanisms cause periodic latency spikes.
- Crowded spectrum bands cause signal interference.



OPC UA

- High resource burden and baseline latency.
- External access requires opening risky firewall ports.
- Heavy security overhead drains device batteries.

Future Strategic Outlook

15

- **Process-aware Intrusion Detection**

Distributed tracing models MQTT communication as expected Petri-net workflows to spot anomalies.

- **Automated Certificate Management**

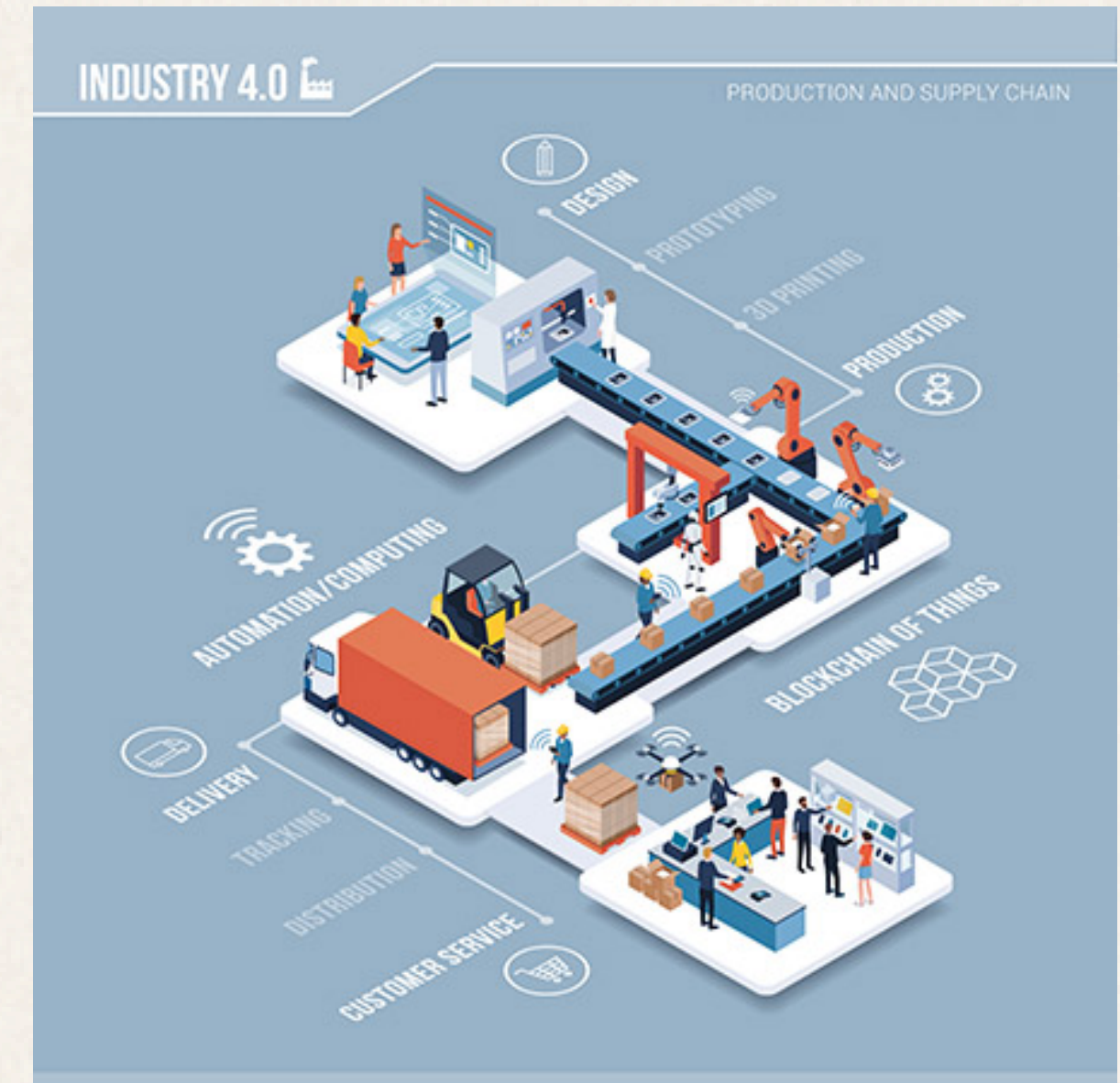
Secure-by-design deployment templates maintain up-to-date cryptographic primitives automatically.

- **Hybrid Protocol Architectures**

Combining OPC UA plant models with lightweight MQTT and CoAP cloud interfaces.

- **Next-Generation Networking**

5G URLLC and Time-Sensitive Networking (TSN) drastically reduce latency and loss.



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Conclusion

- IIoT protocols enable secure Industry 4.0 communication
- MQTT supports fast and lightweight data exchange
- OPC UA provides security and structured data management
- CoAP works well for constrained devices
- Each protocol suits different industrial applications
- Protocol selection depends on system requirements
- Continuous data verification improves system security
- Proper protocol choice ensures reliable operations

References

The information presented today is based on our seminar paper: "IIoT Communication Protocols: A Comparative Analysis of MQTT, OPC UA, and CoAP for Industrial Networks."

Image Reference:

- [1] <https://oringnet.com/en/knowledge-base/what-is-mqtt-and-how-it-works>
- [2] <https://payatu.com/masterclass/iot-security-part-11-introduction-to-coap-protocol-and-security/>
- [3] generated by prompting LLM
- [4] <https://openautomationsoftware.com/open-automation-systems-blog/iiot-edge-computing-vs-cloud-computing/>
- [5] <https://www.twi-global.com/image-library/FAQs/Industry-4.0-diagram.jpg>